

## Part I: Unix Review and File Setup.

If you are on windows, you will *need* to install **git** for this part of the lab. This will allow you to use a shell terminal called “*Git Bash*” locally on your computer. For Mac users, the default “Terminal” application works great.

Consider the following file system on a fellow Berkeley student’s computer. We are looking at the home directory “~”:

```
~  
├── Documents  
│   ├── econ100b  
│   ├── stat134  
│   │   └── syllabus.pdf  
│   │   └── hw  
│   └── data100  
├── Downloads  
│   ├── image.png  
│   ├── download.zip  
│   ├── hw1.pdf  
│   └── f.exe  
...
```

1. Write the relative file path starting at the “Downloads” directory to the “hw” directory in “stat134”.
2. Write the absolute path of “hw1.pdf”.
3. Use the “cp” command to copy the “hw1.pdf” file from Downloads to the “hw” directory in “stat134”. You may use relative or absolute paths.

```
$
```

```
$
```

Now for some sanity checks. Open your computer's Terminal (or Git Bash) and navigate to the home directory "~". This is typically the default directory when you open either application.

For mac users, when you run `pwd` you should see:

```
/Users/[username]
```

For windows running `pwd`:

```
/c/Users/[username]
```

At this point, if you run `ls` you should see `Downloads`, `Documents`, `Desktop`, and more.

4. Create a `stat133` folder within your `~/Documents` directory.

```
$  
$  
$
```

5. Download the `data-in-trees` directory locally. This will be packaged as a `zip` file. Unzip it to your `Downloads` folder. Then copy the new unzipped folder into your new `stat133` directory.

```
$  
$  
$
```

6. *Challenge*: Display the entire tree of contents for `data-in-trees-master` with a single line.<sup>1</sup>

```
$
```

---

<sup>1</sup>Hint: This is known as *recursive* listing. You may reference the [ls manual](#)

## Part II: First Contact with R.

You should have the R App installed on your computer. If not, navigate to [The R Project for Statistical Computing](https://cloud.r-project.org) website and download it from any CRAN mirror (<https://cloud.r-project.org> works just fine).

7. Set the working directory of R session to be the `stats` for your current time at berkeley's folder. For example, if you've been a berkeley student for 1 year, your working directory should be: `"~/Documents/stat133/data-in-trees/stats/one"`.

```
>  
  
>
```

8. Use a `for` loop to fill the vector `"contents"` with the data on each notecard. Your final vector should follow the pattern: `c(song, distance, song, distance, ...)`.<sup>2</sup>

```
> contents <- c()  
  
>  
  
>  
  
>  
  
>  
  
>  
  
>  
  
>
```

9. Fill a matrix `"mat"` with each row corresponding to one student, and each column corresponding to the two questions answered.<sup>3</sup>

```
> mat <-
```

---

<sup>2</sup>Hint: See `readLines()` function. To append to R vectors, use the `c()` function.

<sup>3</sup>Hint: The "byrows" argument in the `matrix()` is useful here.

10. Using a **for loop**, fill the vector `far_distance` with booleans which hold `TRUE` if the student traveled **more than** 1,000 miles over the summer, and `FALSE` otherwise. How many responses hold `TRUE`?<sup>4</sup>

```
> far_distance <- c()
>
>
>
>
>
>
>
>
>
>
>
>
```

11. Using **recycling**, fill and define a vector `far_distance` with booleans which hold `TRUE` if the student traveled more than 1000 miles over the summer, and `FALSE` otherwise. How many responses hold `TRUE` (is it the same is 7)?

```
>
>
>
>
```

---

<sup>4</sup>Hint: Use `as.numeric` to convert character strings into numbers.

12. *Challenge:* Of the different methods you used in the previous two problems, which process do you suspect is faster? Why?

13. Is it possible to fill the second column of `"mat"` with numeric values instead of string values? Why or why not?